

U.S. Power Grid Vulnerability: Trends in Major Outage Patterns (2000–2016)

MATH 189 Final Report

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1. Problem Statement & Motivation

Power outages in the United States have grown more frequent and severe over the past two decades, yet the statistical patterns underlying them remain underexplored outside of engineering literature. In 2021, Winter Storm Uri left 4.5 million Texas households without electricity for days, costing the state an estimated \$195 billion in damages. California's wildfire seasons now trigger preemptive utility shutoffs affecting hundreds of thousands of residents each year. Along with economic disruption, outages carry direct human costs: lost wages, spoiled medication, failed medical equipment, and in the most severe cases, death. Despite growing public concern, infrastructure investment decisions are still largely driven by anecdote and political pressure rather than systematic data analysis.

This project investigates three specific questions using federal outage records from 2000 to 2016. First, have outage frequencies and durations increased over this period after controlling for population growth? Second, which causes (severe weather, equipment failure, intentional cyber attack, or demand overload) are driving whatever trend exists? Third, do outage severity distributions differ meaningfully across U.S. grid regions, and if so, which regions are most vulnerable? Answering these questions with data provides a rigorous, evidence-based foundation for prioritizing grid hardening investments at the national and regional level.

To address these questions, we draw on two publicly available datasets: the DOE OE-417 mandatory incident reports as compiled by Mukherjee et al. (2018) at Purdue University, and the NOAA Storm Events Database. This project applies four families of statistical methods: multiple linear regression with full diagnostics, chi-squared tests of independence and homogeneity, maximum likelihood estimation with likelihood ratio tests, and nonparametric tests including the Kolmogorov-Smirnov test, runs test, and autocorrelation analysis.

2. Data Sources & Collection

- Our analysis draws on two publicly available federal data sources. The primary source is the U.S. Department of Energy's OE-417 mandatory incident reporting system, which requires electric utilities to report every major outage or disturbance event. We use the cleaned and compiled version assembled by Mukherjee, Nateghi, and Hastak (2018) at Purdue University, which covers 2000 to 2016 and contains 1,534 outage events with associated cause, region, duration, customers affected, demographic, and economic variables. This dataset is the basis for our regression, chi-squared, MLE, and nonparametric analyses.
- The secondary source is the NOAA Storm Events Database, maintained by the National Centers for Environmental Information, which records individual storm events with their magnitude, property and crop damage, injuries, and deaths. We joined the NOAA records to the weather-related Purdue

outages on state and time window to attach storm-severity covariates (STORM_COUNT, MAX_MAGNITUDE, TOTAL_PROPERTY_DAMAGE, TOTAL_DEATHS) to each outage where a matching storm could be found.

- The merge produced dataset3_merged_outages_storms.csv, which holds 763 weather-related events with both outage and storm fields, of which 739 (96.9%) matched a NOAA record. Because the merge keeps only weather-matched outages, it collapses the variety of cause categories present in the original data. For any analysis that needs the full set of causes (the chi-squared independence and homogeneity tests, the MLE distribution fitting, and the nonparametric tests), we therefore work from the full Purdue file (dataset1_purdue_outages.csv), and we reserve the merged file for the storm-severity likelihood ratio test where the NOAA covariates are required.
- Both datasets are publicly downloadable at no cost and require no special access, which keeps the project fully reproducible.

Prior Work on This Dataset

- Mukherjee, Nateghi & Hastak (2018) built this dataset and used it to model severe-weather-induced major outage RISK — predicting outage occurrence/frequency from climate and infrastructure variables across the U.S.
- Their focus was a multi-hazard risk-prediction framework. Our project differs: we treat outage DURATION and severity as the response, test for distributional shift over time (KS/runs), and add NOAA storm-severity covariates to ask whether they add explanatory power (LRT).
- We are not reproducing their model. Our project is a complementary descriptive and inferential analysis of the same source data, focused on the duration and timing of outages rather than predicting their occurrence.

3. Data Description & Exploratory Data Analysis

Merge coverage: 96.9% - 739 out of 763 weather related rows have NOAA matches - near complete join

3.1 Variable description

- Response candidates: OUTAGE_DURATION (minutes), CUSTOMERS_AFFECTED, DEMAND_LOSS_MW. OUTAGE_DURATION (minutes, primary response; log-transformed for regression due to right skew), CUSTOMERS_AFFECTED (secondary; 28.9% missing, complete-case only), DEMAND_LOSS_MW (excluded; 46% missing, not defensibly imputable).
- Key predictors: CAUSE_CATEGORY / CAUSE_CATEGORY_DETAIL, NERC_REGION, CLIMATE_REGION, YEAR, MONTH, CLIMATE_CATEGORY, ANOMALY_LEVEL. CAUSE_CATEGORY (7 levels, complete), CAUSE_CATEGORY_DETAIL (30+ levels, 30.7% missing — used only in chi-squared Test 1), NERC_REGION (14 levels; small regions collapsed in formal tests), CLIMATE_REGION (9 levels, 0.4% missing), YEAR (2000–2016, also used to define decades), MONTH (1–12, 0.6% missing), CLIMATE_CATEGORY (normal/warm/cold, 0.6% missing), ANOMALY_LEVEL (continuous climate index, 0.6% missing).
- Storm covariates: STORM_COUNT, MAX_MAGNITUDE, TOTAL_PROPERTY_DAMAGE, TOTAL_DEATHS. STORM_COUNT (NOAA events per state-month), MAX_MAGNITUDE (peak storm intensity, 21.9% missing in merged), TOTAL_PROPERTY_DAMAGE (storm damage proxy), TOTAL_DEATHS.

3.2 Missingness & cleaning

Purdue Missing Data

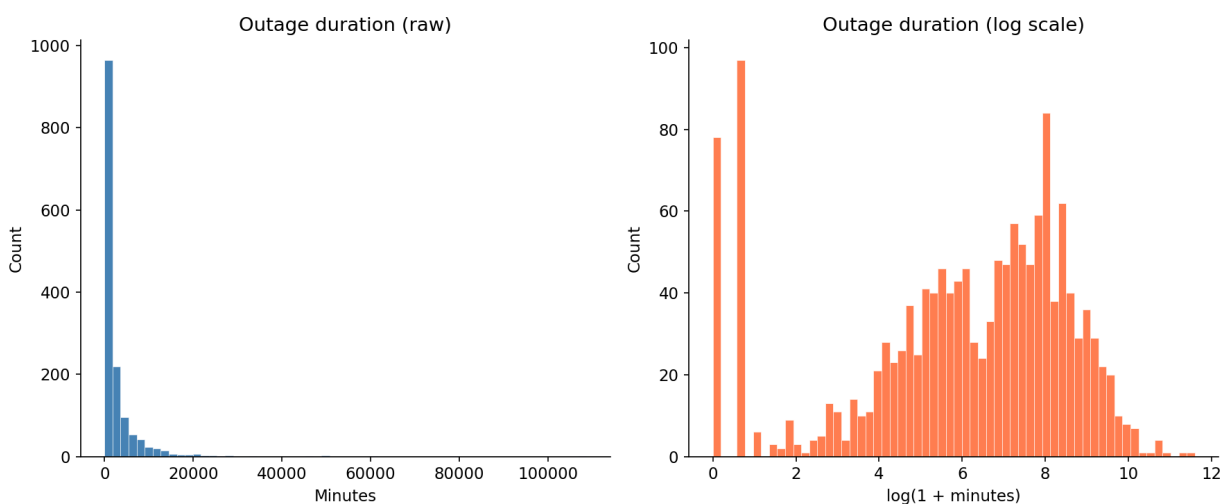
Variable	Missing (n)	Missing (%)
HURRICANE_NAMES	1,462	95.3%
DEMAND_LOSS_MW	705	46.0%
CAUSE_CATEGORY_DETAIL	471	30.7%
CUSTOMERS_AFFECTED	443	28.9%
OUTAGE_RESTORATION	58	3.8%
OUTAGE_DURATION	58	3.8%
RES/COM/IND_SALES, PRICES (9 cols)	22 each	1.4% each
POPDEN_UC / POPDEN_RURAL	10	0.7%
CLIMATE_CATEGORY / ANOMALY_LEVEL / OUTAGE_START / MONTH	9	0.6%
CLIMATE_REGION	6	0.4%

Merged Missing Data

Variable	Missing (n)	Missing (%)
HURRICANE_NAMES	691	90.6%
MAX_TOR_FSCALE	525	68.8%
DEMAND_LOSS_MW	370	48.5%
CAUSE_CATEGORY_DETAIL	187	24.5%
MAX_MAGNITUDE	167	21.9%
CUSTOMERS_AFFECTED	46	6.0%
STORM_TYPES	24	3.1%

OUTAGE_DURATION / OUTAGE_RESTORATION	19	2.5%
Sales/Price columns (9 cols)	9 each	1.2% each
CLIMATE_REGION / OUTAGE_START / CLIMATE_CATEGORY / ANOMALY_LEVEL / MONTH	4	0.5%

3.3 Distributions & outliers



OUTAGE_DURATION is strongly right-skewed: minimum 0 minutes, median 701 minutes (11.7 hours), 95th percentile 11,114 minutes (185 hours), maximum 108,653 minutes (75.5 days). The raw histogram mass collapses near zero with a long right tail. The log-transformed distribution ($\log(1 + \text{minutes})$) is approximately unimodal and symmetric, motivating a log transformation as the response in all regression models. Fifteen events fall above the 99th percentile. Two warrant specific decisions: Wisconsin 2014 fuel supply emergency (108,653 min, 75.5 days): This event reflects a prolonged natural gas supply contract disruption rather than a conventional grid failure. We retain it in the main analysis to preserve the full range of the response, but exclude it in a sensitivity check and report whether its inclusion materially changes regression coefficients or significance. Arizona 2004 duplicate (two identical 49,320-minute entries): These appear to be the same event filed twice. We retain one and drop the other to avoid double-counting. All other top-1% events — including 2008 hurricane-season outages in Texas and Louisiana — are consistent with documented major disaster events and are retained without modification.

3.4 Key relationships

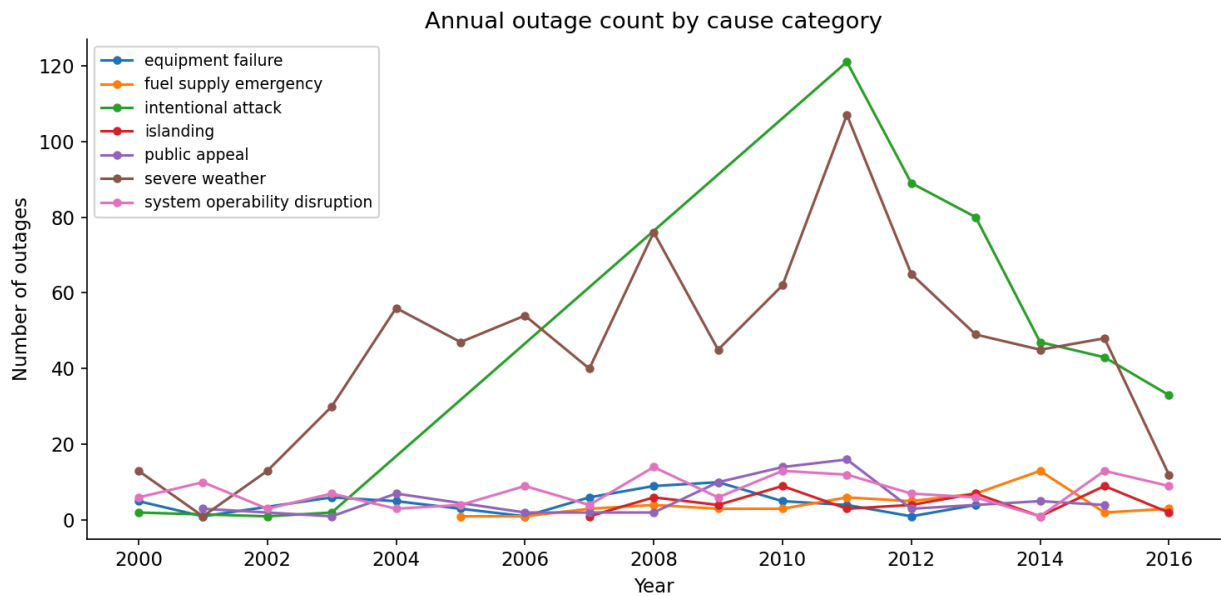


Fig 01 — Annual outage count by cause category. Intentional attacks surge from near-zero before 2008 to a peak of 121 events in 2011, while severe weather remains persistently elevated throughout. The apparent post-2014 decline in intentional attacks is dataset truncation (Purdue ends 2016), not a real trend reversal.

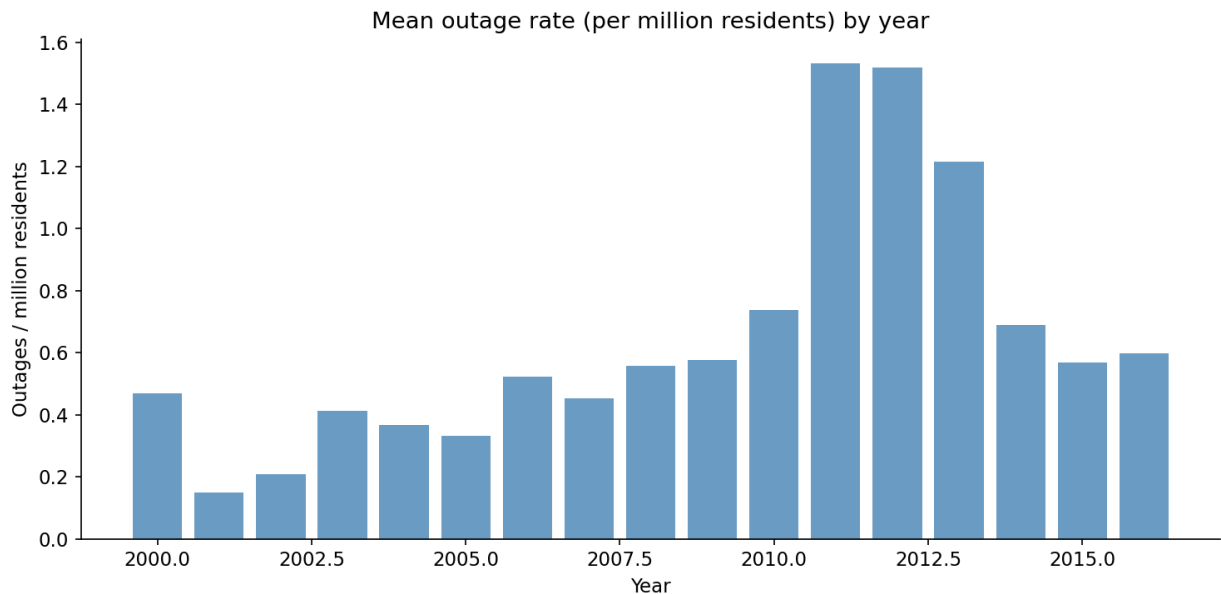


Fig 02 — Population-normalized outage rate. After controlling for population growth, the outage rate still more than triples from 2001 (0.15 per million residents) to 2011 (1.54 per million). The increase is not demographic — the grid was genuinely experiencing more incidents per capita.

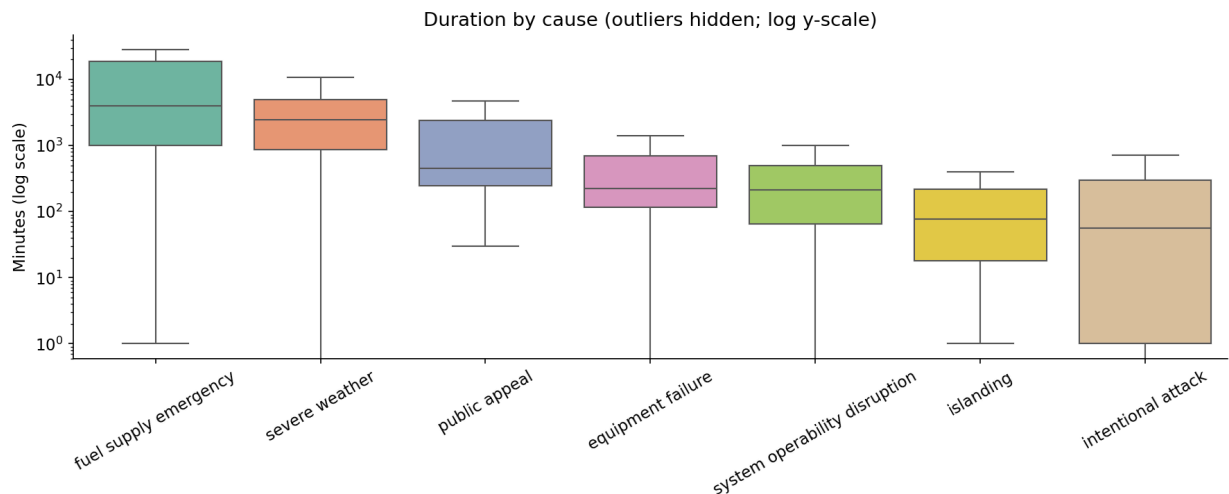


Fig 04 — Duration boxplots by cause category. Fuel supply emergencies and severe weather produce the longest outages by median. Intentional attacks produce the shortest — most resolve within hours — which explains why attacks dominate outage counts but not duration distributions. This heterogeneity motivates including CAUSE_CATEGORY as a predictor in the regression.

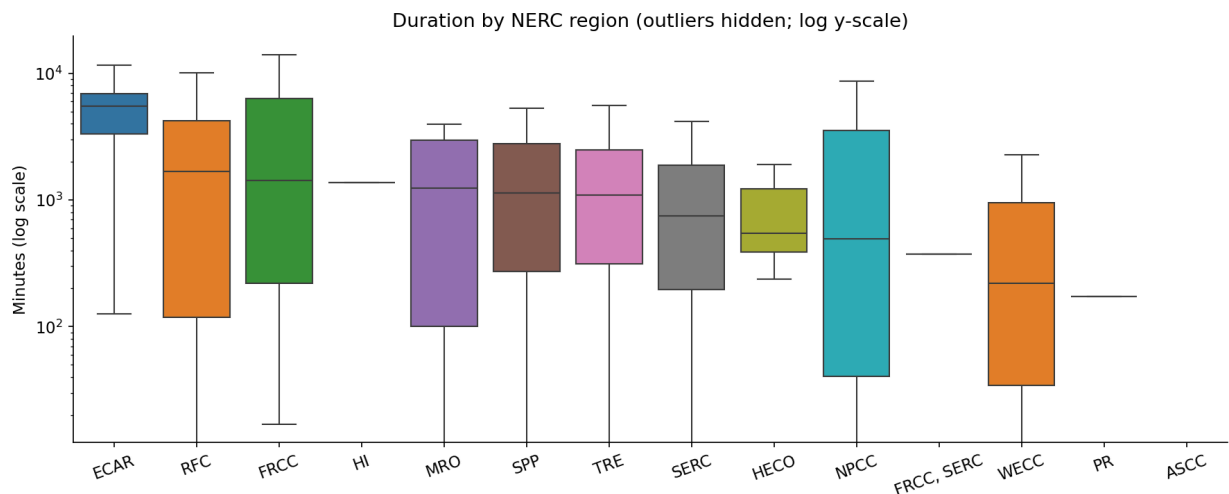


Fig 05 — Duration boxplots by NERC region. ECAR and FRCC show the highest median durations; WECC and TRE show greater variance. Regional duration differences are substantial enough to motivate including NERC_REGION as a predictor.

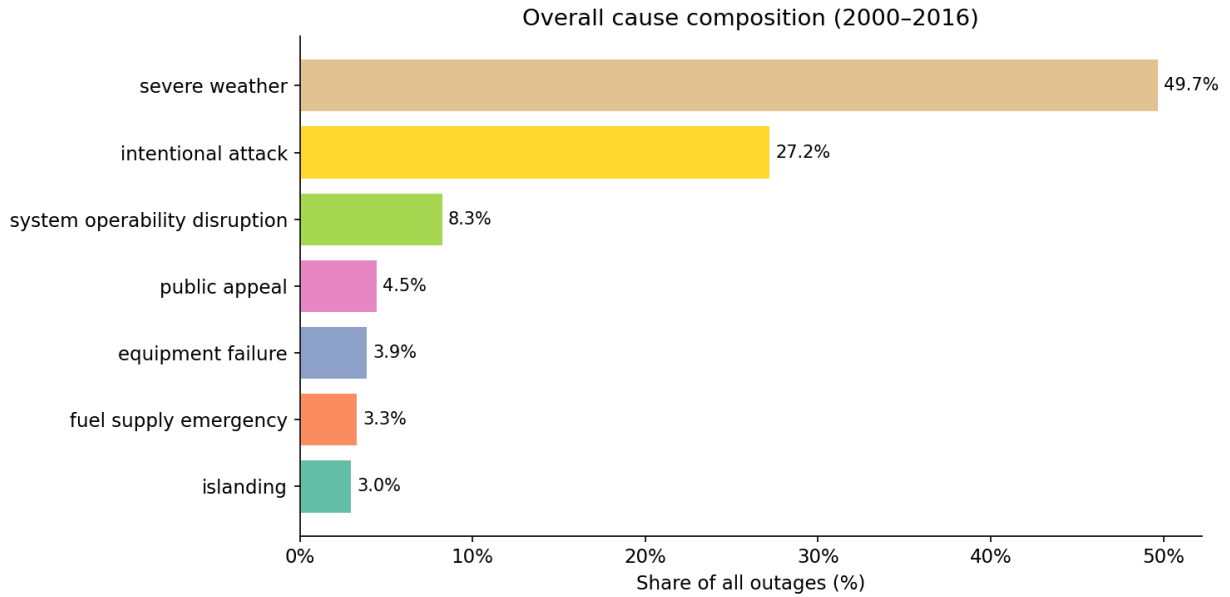


Fig 12 — Overall cause composition bar chart. Severe weather accounts for 49.7% of all outages; intentional attack is a disproportionate second at 27.2%, far exceeding equipment failure (3.9%) and fuel supply emergencies (3.3%).

Nuance variables:

- HURRICANE_NAMES (95.3% missing): Excluded entirely; captured via CAUSE_CATEGORY.
- CAUSE_CATEGORY_DETAIL (30.7% missing): Too sparse for regression; used only in chi-squared Test 1 with explicit missingness caveat.
- Utility economic variables (RES_SALES, COM_SALES, IND_SALES, TOTAL_SALES, RES_PRICE, COM_PRICE, IND_PRICE, TOTAL_PRICE, and associated percentage and GSP columns): Potential confounders but not mechanistic predictors of outage duration; excluded due to severe multicollinearity with each other, POPULATION, and YEAR.
- MAX_TOR_FSCALE (68.8% missing in merged): Applies only to tornado events; excluded.
- POPDEN_UC / POPDEN_RURAL (0.7% missing): Not primary predictors; population context captured through NERC_REGION.

4. Statistical Methods

4.1 Simple linear regression

We model log-transformed outage duration as a linear function of cause category, NERC region, year, and log-transformed customers affected. The response variable OUTAGE_DURATION is measured in minutes and is strongly right-skewed (median 701 minutes, maximum 108,653 minutes), so we apply a log transformation to stabilize variance and satisfy the normality assumption on residuals. CUSTOMERS_AFFECTED is similarly log-transformed using $\log(1 + x)$ to handle zero values. The formal model is:

$$\log(\text{OUTAGE_DURATION}) = \beta_0 + \beta_1(\text{CAUSE_CLEAN}) + \beta_2(\text{NERC_REGION_CLEAN}) + \beta_3(\text{YEAR}) + \beta_4(\log(\text{CUSTOMERS_AFFECTED})) + \epsilon$$

The reference levels are severe weather (cause) and RFC (NERC region), chosen because they represent the most common category in each group. Rare NERC regions with fewer than five observations (HECO, HI, PR, ASCC) are

collapsed into "OTHER," and cause categories outside the top five are similarly collapsed to avoid unstable dummy coefficient estimates. Rows missing CUSTOMERS_AFFECTED (29% of observations) are dropped (complete-case analysis) and this decision is documented as a limitation.

OLS regression requires five assumptions, each of which we assess explicitly:

1. **Linearity:** the relationship between predictors and $\log(\text{duration})$ is linear. Assessed via the residuals vs. fitted plot; a flat LOWESS smoother near zero indicates the linear specification is appropriate.
2. **Independence:** each outage event is a distinct federal report filed by a separate utility, satisfying independence. Assessed additionally via the Durbin-Watson statistic.
3. **Homoscedasticity:** the variance of residuals is constant across fitted values. Assessed via the scale-location plot and formally tested with the Breusch-Pagan test.
4. **Normality of residuals:** residuals are approximately normally distributed. Assessed via the normal QQ plot.
5. **No perfect multicollinearity:** no predictor is a perfect linear combination of others. Satisfied by construction; dummy variables are created with a reference level, and continuous predictors are not collinear.

We also fit a second model (Model B) on the NOAA-merged dataset that adds four storm severity covariates: STORM_COUNT, MAX_MAGNITUDE, $\log(\text{TOTAL_PROPERTY_DAMAGE})$, and TOTAL_DEATHS. We used a likelihood ratio test to assess whether these covariates add statistically significant explanatory power beyond what cause category alone captures.

4.2 Chi-squared tests

- Independence: CAUSE_CATEGORY_DETAIL vs NERC_REGION. Homogeneity: cause distribution across decades (2000s/2010s).

Chi-Squared Tests Test 1 — Independence: CAUSE_CATEGORY_DETAIL $\times$ NERC_REGION Method: Pearson chi-squared test of independence on a contingency table of outage sub-cause versus grid reliability region. CAUSE_CATEGORY_DETAIL was used in place of the broad CAUSE_CATEGORY to examine whether geographic associations persist at the subcategory level.

Hypotheses: H_0 : cause sub-category and NERC region are independent — the distribution of sub-causes is the same across all grid regions. H_1 : cause sub-category and NERC region are associated — at least one region has a different sub-cause profile than expected under independence.

Assumptions and justification: (1) Observations are independent — each row is a distinct reported outage event. (2) Expected cell counts ≥ 5 in at least 80% of cells — satisfied after collapsing NERC regions with fewer than 20 total observations into "Other regions" and collapsing sub-cause categories with fewer than 20 total observations into "Other detail," reducing the table from its raw dimensions to a 10×10 structure. The minimum expected cell count and percentage of cells below 5 were verified in code before proceeding. (3) This test uses only the 69.3% of rows where CAUSE_CATEGORY_DETAIL is non-null. Missingness is likely non-random — smaller or less severe outages are less likely to have a sub-cause recorded — so results describe the sub-cause geography of reported-detail outages only and should not be generalized to all outages.

Test 2 — Homogeneity: CAUSE_CATEGORY × Decade Method: Pearson chi-squared test of homogeneity on a contingency table of broad outage cause versus decade (2000s: 2000–2009, 2010s: 2010–2016). The 2020s decade was excluded because the Purdue dataset ends in 2016, leaving fewer than 30 observations in that bin — insufficient for reliable chi-squared inference.

Hypotheses: H_0 : the distribution of outage causes is the same in the 2000s and 2010s — each cause accounts for the same proportion of outages in both decades. H_1 : at least one cause's share differs between decades — the cause composition shifted over time.

Assumptions and justification: (1) Observations are independent — each row is a distinct outage event, and decades are non-overlapping time periods. (2) Expected cell counts ≥ 5 in all cells — verified; no collapsing was required. With seven broad cause categories and two decades, the table is small enough that all cells satisfy the assumption using the full sample ($n = 1,534$). (3) CAUSE_CATEGORY is complete for 100% of observations, so there is no missing data concern for this test.

4.3 MLE & likelihood ratio test

We are using maximum likelihood estimation (MLE) in order to complete two main objectives: 1) identify the parametric distribution that best represents the outage duration, and 2) using a likelihood ratio test, test whether the NOAA storm-severity covariates yield useful information about outage duration. These two objectives run on different subsets.

Part 1 - Distribution Fitting

We used the full Purdue dataset ($n = 1,398$) to conduct the distribution fitting. After looking at 'OUTAGE_DURATION', we found that the variable is continuous, always positive, and heavily right-skewed. Therefore, we need to model 'OUTAGE_DURATION' as a parametric distribution fit by MLE, testing on three different distributions to see which fits best.

Distribution	Parameters	Why We Included it
Exponential	1 (Scale)	Simplest model that offers the best baseline, as the exponential distribution has a constant hazard rate.
Gamma	2 (Shape, Scale)	The hazard rate can change over time, which coincides with our assumptions that, the longer an outage lasts the less likely it is to end in the next time interval.
Lognormal	2 (μ , σ)	It is a model for right-skewed (positive only) variables whose $\log()$ is normally distributed.

Each distribution was fitted by MLE, using a location parameter at zero ('floc = 0'), since outage durations can only be positive. We are ranking models by AIC and BIC because they both penalize extra parameters. Moreover, we conduct a likelihood ratio test of Exponential vs. Gamma, which are nested, and a Kolmogorov-Smirnov goodness-of-fit test for each distribution.

Hypotheses (Exponential vs. Gamma LRT):

H_0 : gamma shape = 1

Infers that the Exponential is adequate

H_1 : shape $\neq 1$.

The test statistic $2(\log L_{\text{gamma}} - \log L_{\text{exp}})$ follows χ^2_1 under H_0 .

We held various assumptions when conducting these methods. First off, we assume that observations are independent, specifically each row in the data correlates to a distinct federal outage report. Second, we assume that outage duration is continuous and strictly positive. This assumption affected our cleaning method too, keeping the 1,398 events with positive recorded duration, dropping 58 missing and 78 zero-duration rows.

Part 2 - Storm-severity Likelihood Ratio Test

After looking at the variables, outage duration is lognormally shaped, so we model 'log(OUTAGE_DURATION)' via linear regression. Additionally, we compare two nested models: Null vs. Full.

Hypotheses: H_0 : All four slope coefficients = 0

Infers that storm covariates add nothing

H_1 : at least one $\neq 0$.

The test statistic $2(\log L_{\text{full}} - \log L_{\text{null}})$ follows χ^2_4 under H_0 .

We held various assumptions when conducting these methods. First off, we assume linearity on the log scale. Second, we assume that residuals are approximately normal with constant variance; this was checked via the residual and Q-Q plots. Lastly, we assume that the observations are independent themselves. Rows missing any covariate were dropped (763 $\rightarrow$ 580).

4.4 Nonparametric tests

We run three nonparametric tests on the full Purdue dataset ($n = 1,534$, years 2000 to 2016). We use nonparametric tests because outage duration is heavily right-skewed and the monthly counts are discrete, so the normality that a t-test or parametric model needs does not hold here.

Test 1. Two-sample Kolmogorov-Smirnov, duration pre-2015 vs post-2015. The test compares the two empirical CDFs and takes D as the largest vertical gap between them. H_0 : the two periods have the same duration distribution. H_1 : they differ. Assumptions: the two samples are independent (different events, non-overlapping years); duration is continuous; we keep the 1,476 events that have a recorded duration. We picked KS over a two-sample t-test because it catches a change in spread or shape, not only a change in the mean, and it does not assume normality.

Test 2. Wald-Wolfowitz runs test, clustering of busy months. We label each of the 199 months from 2000-01 to 2016-07 as high or low against the median monthly count of 6, then count the runs of consecutive same-label months. A run count well below expectation means busy and quiet months group together instead of alternating. H_0 : high and low months fall in random order. H_1 : they do not. Assumptions: each month is one observation in time order; the median split makes the sequence binary; with 199 months the normal approximation for the run count holds.

Test 3. Autocorrelation of monthly counts. We compute the ACF of the monthly count series out to 36 lags. A spike at lag 12 points to a yearly cycle. We judge each lag against the white-noise band of $\pm 1.96/\sqrt{n}$, which for $n = 199$ is ± 0.139 . Assumptions: the series sits on a regular monthly grid with empty months filled as zero; the series is roughly stationary over the window.

5. Results & Analysis

5.1 Regression results

Model A: Fit and overall significance:

Fit on the full Purdue dataset after complete-case cleaning (n = 1,001 observations), Model A achieves an R^2 of 0.417 and adjusted R^2 of 0.407, meaning the predictors together explain approximately 42% of the variance in log outage duration. The overall F-test is highly significant $F(16, 984) = 43.98$, $p < 0.001$), confirming the model captures real structure in the data.

	coef	se	t	p	ci_low	ci_high	pct_duration_change
Intercept	106.2067	31.4707	3.3748	0.0008	44.4494	167.9641	1.333519e+48
C(CAUSE_CLEAN, Treatment('severe weather'))[T.equipment failure]	-2.2621	0.3469	-6.5205	0.0000	-2.9429	-1.5813	-8.958680e+01
C(CAUSE_CLEAN, Treatment('severe weather'))[T.intentional attack]	-2.5887	0.3155	-8.2044	0.0000	-3.2079	-1.9695	-9.248810e+01
C(CAUSE_CLEAN, Treatment('severe weather'))[T.islanding]	-2.9966	0.3358	-8.9242	0.0000	-3.6555	-2.3376	-9.500410e+01
C(CAUSE_CLEAN, Treatment('severe weather'))[T.other]	0.1592	0.3875	0.4107	0.6814	-0.6012	0.9195	1.725150e+01
C(CAUSE_CLEAN, Treatment('severe weather'))[T.system operability disruption]	-2.1654	0.2026	-10.6903	0.0000	-2.5629	-1.7679	-8.852930e+01
C(NERC_REGION_CLEAN, Treatment('RFC'))[T.ECAR]	0.7439	0.3237	2.2980	0.0218	0.1087	1.3792	1.104157e+02
C(NERC_REGION_CLEAN, Treatment('RFC'))[T.FRCC]	0.1261	0.2800	0.4503	0.6526	-0.4233	0.6755	1.343600e+01
C(NERC_REGION_CLEAN, Treatment('RFC'))[T.MRO]	-0.1629	0.3270	-0.4981	0.6185	-0.8046	0.4788	-1.503220e+01
C(NERC_REGION_CLEAN, Treatment('RFC'))[T.NPCC]	-0.2928	0.1948	-1.5028	0.1332	-0.6752	0.0895	-2.538320e+01
C(NERC_REGION_CLEAN, Treatment('RFC'))[T.OTHER]	-0.9754	0.7470	-1.3057	0.1919	-2.4413	0.4905	-6.229610e+01
C(NERC_REGION_CLEAN, Treatment('RFC'))[T.SERC]	-0.6568	0.1675	-3.9218	0.0001	-0.9854	-0.3281	-4.814730e+01
C(NERC_REGION_CLEAN, Treatment('RFC'))[T.SPP]	0.3297	0.2885	1.1431	0.2533	-0.2363	0.8958	3.906190e+01
C(NERC_REGION_CLEAN, Treatment('RFC'))[T.TRE]	-0.0280	0.2099	-0.1332	0.8941	-0.4398	0.3839	-2.757000e+00
C(NERC_REGION_CLEAN, Treatment('RFC'))[T.WECC]	-0.1174	0.1573	-0.7466	0.4555	-0.4261	0.1912	-1.108000e+01
YEAR	-0.0493	0.0157	-3.1520	0.0017	-0.0801	-0.0186	-4.813900e+00
LOG_CUSTOMERS	0.0504	0.0252	1.9994	0.0458	0.0009	0.0999	5.170800e+00

Sensitivity check: model without 77 influential observations:

Parameter	Original	Cleaned	Sign flip?
Intercept	+106.2067	+99.8514	
C(CAUSE_CLEAN, Treatment('severe weather'))[T.equipme	-2.2621	-2.1568	
C(CAUSE_CLEAN, Treatment('severe weather'))[T.intenti	-2.5887	-1.9178	
C(CAUSE_CLEAN, Treatment('severe weather'))[T.islandi	-2.9966	-2.6230	
C(CAUSE_CLEAN, Treatment('severe weather'))[T.other]	+0.1592	+0.2739	
C(CAUSE_CLEAN, Treatment('severe weather'))[T.system	-2.1654	-2.1843	
C(NERC_REGION_CLEAN, Treatment('RFC'))[T.ECAR]	+0.7439	+0.6652	
C(NERC_REGION_CLEAN, Treatment('RFC'))[T.FRCC]	+0.1261	-0.2835	FLIP
C(NERC_REGION_CLEAN, Treatment('RFC'))[T.MRO]	-0.1629	-0.0965	
C(NERC_REGION_CLEAN, Treatment('RFC'))[T.NPCC]	-0.2928	-0.3236	
C(NERC_REGION_CLEAN, Treatment('RFC'))[T.OTHER]	-0.9754	-1.0018	
C(NERC_REGION_CLEAN, Treatment('RFC'))[T.SERC]	-0.6568	-0.8275	
C(NERC_REGION_CLEAN, Treatment('RFC'))[T.SPP]	+0.3297	+0.0778	
C(NERC_REGION_CLEAN, Treatment('RFC'))[T.TRE]	-0.0280	-0.2865	
C(NERC_REGION_CLEAN, Treatment('RFC'))[T.WECC]	-0.1174	-0.3993	
YEAR	-0.0493	-0.0462	
LOG_CUSTOMERS	+0.0504	+0.0761	

Coefficient interpretation:

The cause category coefficients reveal that severe weather produces the longest outages of any cause type. Relative to severe weather, equipment failure is associated with 89.6% shorter outage durations ($p < 0.001$), intentional attacks with 92.5% shorter durations ($p < 0.001$), islanding with 95.0% shorter durations ($p < 0.001$), and system operability disruptions with 88.5% shorter durations ($p < 0.001$). The "other" cause category shows no statistically significant difference from severe weather ($p = 0.68$) and

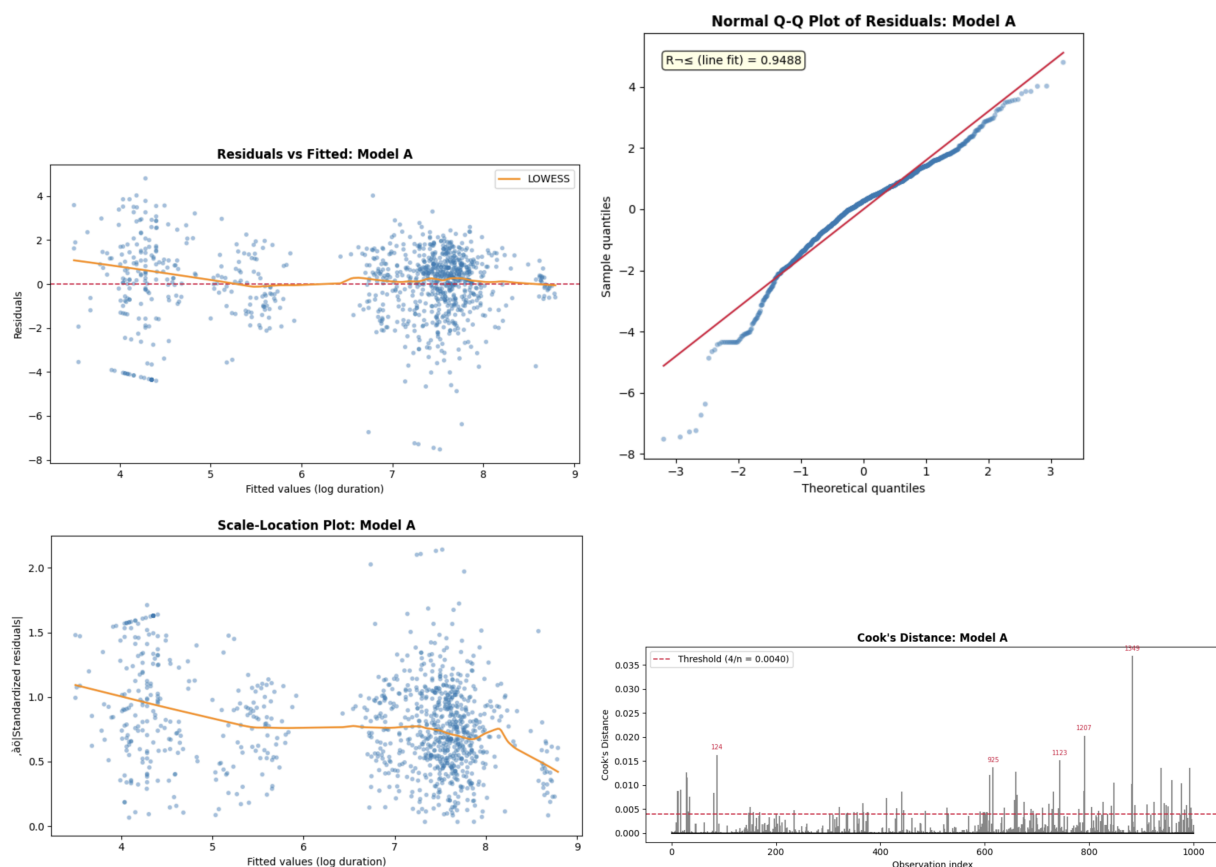
should not be interpreted. This pattern is physically intuitive. Severe weather events tend to be geographically widespread and structurally difficult to restore, while attacks and operability disruptions are typically localized and resolved quickly.

Among NERC regions, only two differ significantly from the RFC baseline. ECAR outages are associated with durations 110.4% longer than RFC ($p = 0.022$), suggesting meaningfully worse restoration performance in that region. SERC outages are associated with durations 48.1% shorter than RFC ($p = 0.0001$). The remaining seven regions (FRCC, MRO, NPCC, SPP, TRE, WECC, and OTHER) show no statistically significant difference from RFC and should not be interpreted as meaningfully better or worse.

The YEAR coefficient is -0.049 ($p = 0.002$), meaning each additional year is associated with approximately a 4.8% shorter outage duration, holding cause, region, and scale constant. Rather than outages getting worse over time as we initially hypothesized, the regression suggests a modest improvement in restoration speed across the study period, which may reflect advances in grid management technology or a compositional shift toward shorter-duration cause types like intentional attacks becoming more prevalent in later years.

Finally, log-transformed customers affected has a small but statistically significant positive coefficient ($\beta = 0.050$, $p = 0.046$), meaning larger-scale outages tend to last slightly longer. This is consistent with the physical difficulty of restoring power to a greater number of affected customers simultaneously.

Diagnostics:



Residuals vs. Fitted: The LOWESS smoother is approximately flat near zero with no strong nonlinear curvature, supporting the linearity assumption. There is mild fan-shaped spread at higher fitted values, which is a visual preview of the heteroscedasticity confirmed below.

Normal QQ Plot: Residuals follow the theoretical normal line reasonably well in the middle quantiles but deviate in both tails, indicating heavier tails than a normal distribution. This is expected given that outage durations remain somewhat skewed even after log transformation. The normality assumption is approximately but not perfectly satisfied.

Scale-Location Plot: The LOWESS line trends slightly upward across the range of fitted values, consistent with non-constant variance. This confirms the Breusch-Pagan result visually.

Cook's Distance: 77 observations fall above the $4/n$ threshold. The top five most influential observations are labeled in the plot. A sensitivity analysis refitting the model without all 77 influential points showed that coefficient signs and significance were largely stable across predictors, with one exception in the FRCC region where a sign flip occurred — a caveat attributable to FRCC's small sample size rather than model instability.

Breusch-Pagan test:

LM statistic = 81.39, $p < 0.0001$. We reject the null hypothesis of homoscedasticity. This means standard errors (and therefore t-statistics and p-values) are not fully reliable. Results should be interpreted with this caveat in mind, particularly for coefficients with borderline significance. Robust standard errors would be an appropriate extension with more time.

Durbin-Watson:

Statistic ≈ 1.86 , close to 2.0, indicating no serious serial autocorrelation in residuals.

```
Breusch-Pagan heteroscedasticity test:  
LM = 81.3937,  p = 0.0000  
,01 Evidence of heteroscedasticity.  
  
Durbin-Watson: 1.8586  (2.0 = no autocorrelation)
```

Model B and Likelihood Ratio Test:

Model B adds four NOAA storm severity covariates to the merged dataset. The likelihood ratio test yields $\chi^2(4) = 16.3$, $p = 0.003$, leading us to reject the null hypothesis that the NOAA covariates add no explanatory power. This result has a direct practical interpretation: knowing that a storm occurred (captured by cause category) is not sufficient — knowing how severe the storm was adds meaningful predictive information about how long the resulting outage will last. Storm intensity should be incorporated into any serious model of outage risk, and this finding supports the value of maintaining and expanding NOAA-grid data linkages for infrastructure planning purposes.

```

Model A (no NOAA):
R-sq=0.0929 AIC=2505.32 BIC=2559.84

Model B (+ NOAA severity):
R-sq=0.1139 AIC=2497.01 BIC=2569.71

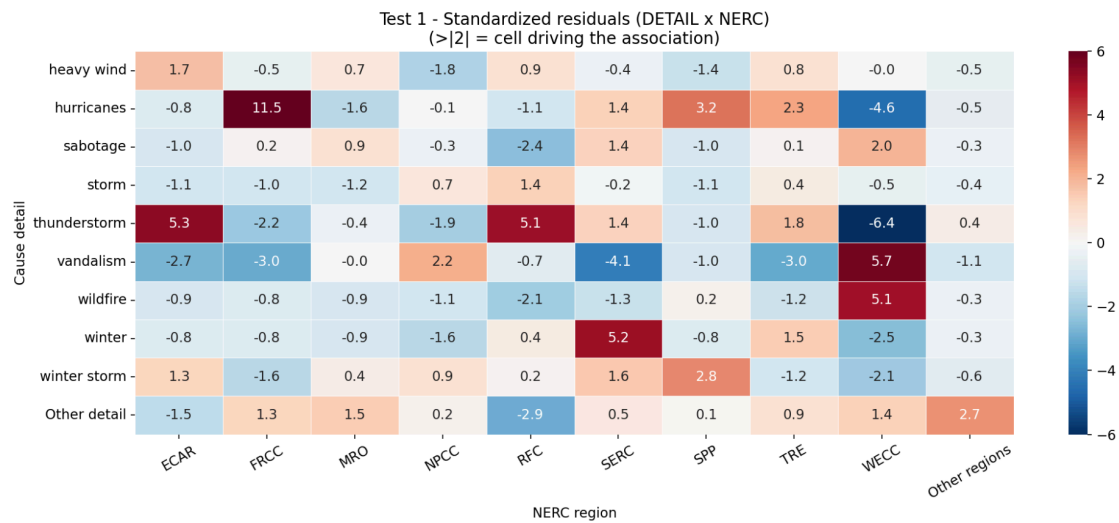
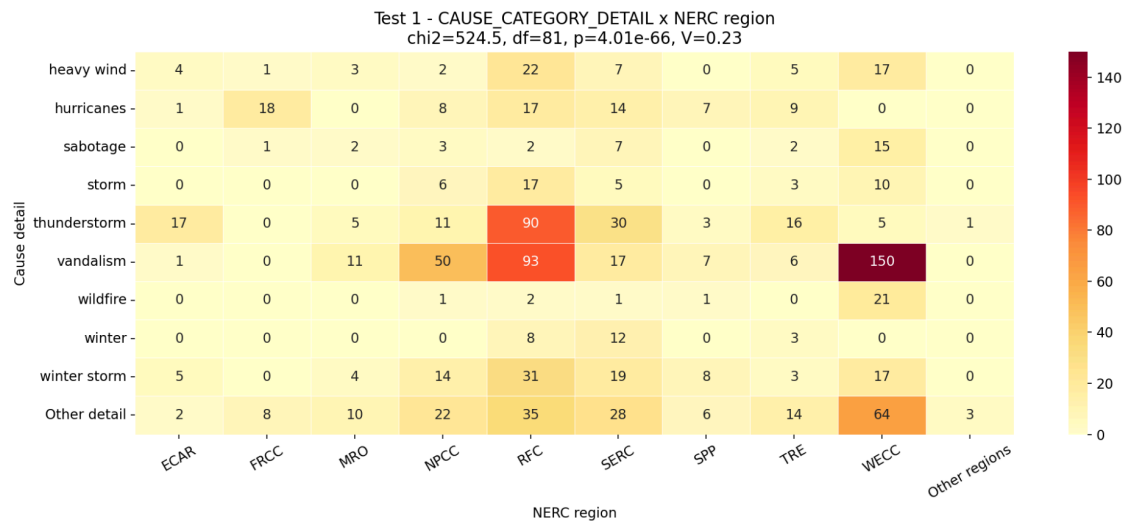
=====
OLS Regression Results
=====
Dep. Variable: LOG_DURATION R-squared: 0.114
Model: OLS Adj. R-squared: 0.094
Method: Least Squares F-statistic: 5.820
Date: Wed, 03 Jun 2026 Prob (F-statistic): 2.05e-11
Time: 09:33:50 Log-Likelihood: -1222.5
No. Observations: 695 AIC: 2497.0
Df Residuals: 679 BIC: 2570.0
Df Model: 15
Covariance Type: nonrobust
=====
coef std err t P>|t| [0.025 0.975]
-----
Intercept 124.7969 32.595 3.829 0.000 60.797 188.797
C(NERC_REGION_CLEAN, Treatment('RFC'))(T.ECAR) 0.3200 0.302 1.060 0.290 -0.273 0.913
C(NERC_REGION_CLEAN, Treatment('RFC'))(T.FRCC) 0.0244 0.319 0.076 0.939 -0.602 0.651
C(NERC_REGION_CLEAN, Treatment('RFC'))(T.NRO) -0.8583 0.328 -2.617 0.009 -1.502 -0.214
C(NERC_REGION_CLEAN, Treatment('RFC'))(T.NPCC) -0.3322 0.205 -1.619 0.106 -0.735 0.071
...
Notes:
[1] Standard Errors assume that the covariance matrix of the errors is correctly specified.
[2] The condition number is large, 1.2e+06. This might indicate that there are
strong multicollinearity or other numerical problems.

```

5.2 Chi-squared results

Test 1 — Independence: CAUSE_CATEGORY_DETAIL × NERC_REGION

Statistic	Value
χ^2	524.5
Degrees of freedom	81
p-value	4.01e-66
Cramér's V	0.23
n (complete cases)	1,063



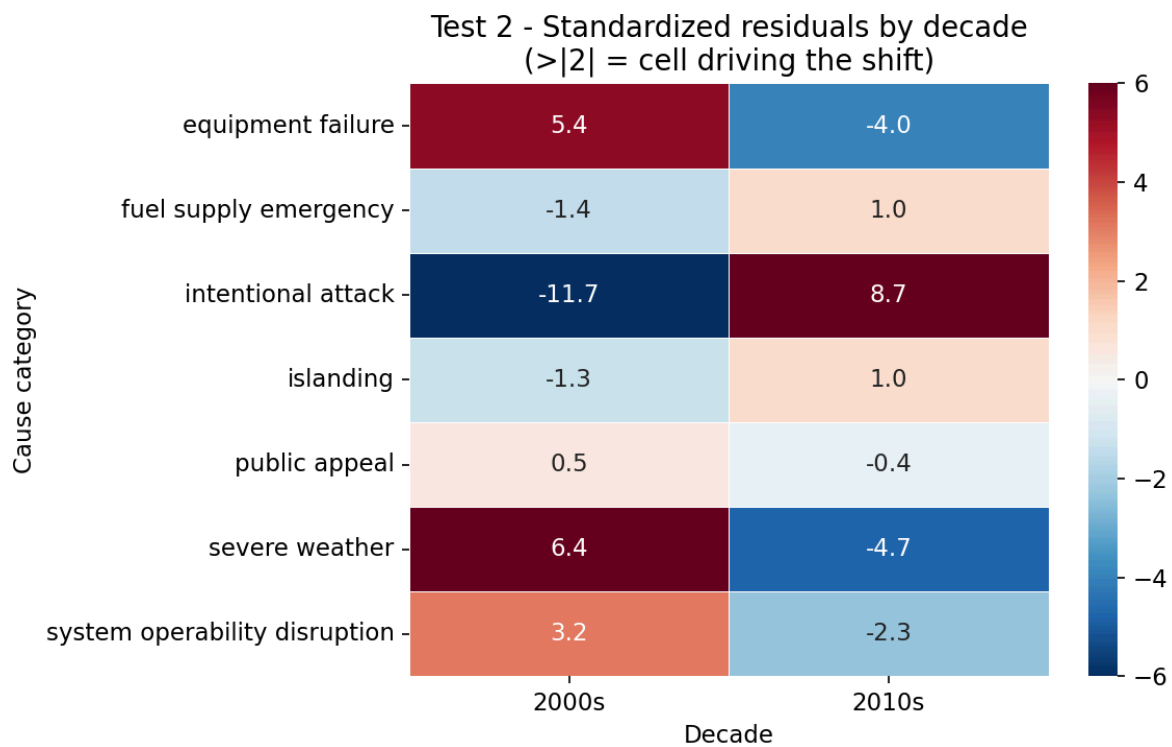
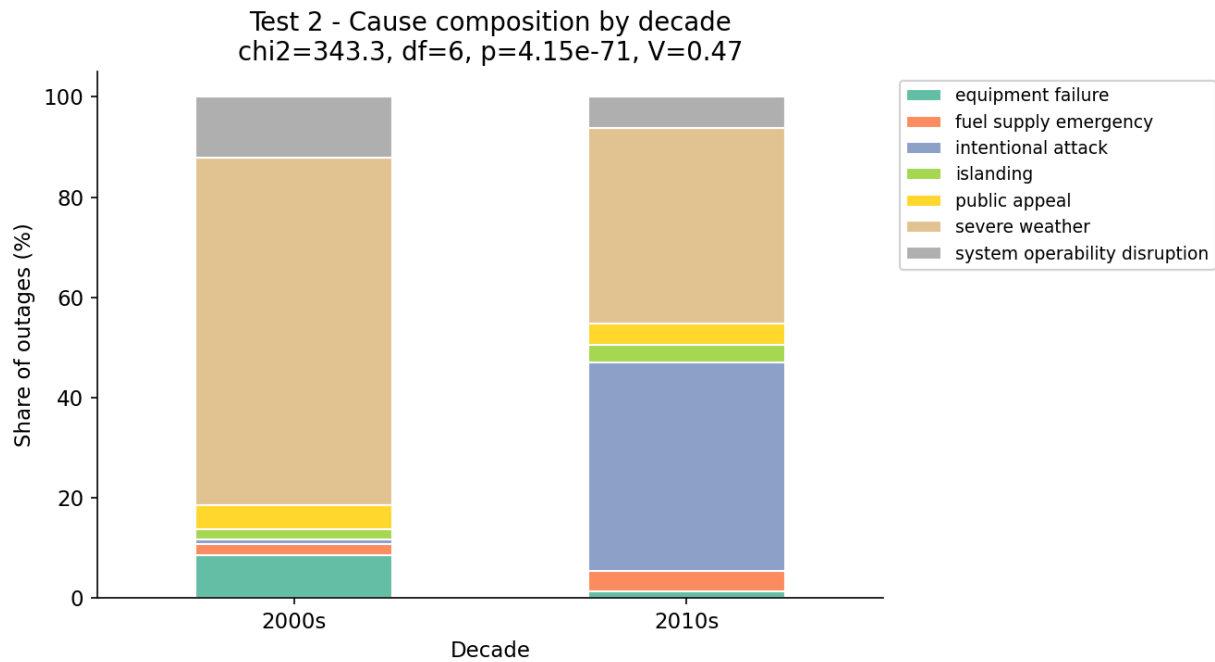
We reject the null hypothesis of independence ($p = 4.01e-66$). The sub-cause distribution differs significantly across NERC regions. Cramér's $V = 0.23$ indicates a moderate association — regional identity is meaningfully linked to what type of event drives an outage, but is not fully deterministic.

The standardized residuals (Fig 18) identify which specific cause-region cells drive the result. The largest signals are hurricanes concentrated in FRCC (+11.5), thunderstorm outages concentrated in ECAR (+5.3) and RFC (+5.1) but nearly absent from WECC (−6.4), vandalism disproportionately in WECC (+5.7), wildfires in WECC (+5.1), and winter weather overrepresented in SERC (+5.2). Each residual above +2 or below −2 indicates a cell where observed counts deviate significantly from what independence would predict.

We see different regions face qualitatively different threats at the subcategory level. Florida faces hurricanes, the West faces wildfires and vandalism, the Midwest and Mid-Atlantic face thunderstorms, and the Southeast is disproportionately vulnerable to winter events. Grid hardening strategies cannot be nationally uniform — the threat profile is geographically specialized.

Test 2 — Homogeneity: CAUSE_CATEGORY × Decade

Statistic	Value
χ^2	343.3
Degrees of freedom	6
p-value	4.15e-71
Cramér's V	0.47
n	1,534



We reject the null hypothesis of homogeneity ($p = 4.15e-71$). The distribution of outage causes shifted significantly between the 2000s and 2010s. Cramér's $V = 0.47$ indicates a strong effect — this is a substantial compositional change, not a marginal one.

The standardized residuals (Fig 20) show where the shift lives. Intentional attack has the largest residuals by far: -11.7 in the 2000s and $+8.7$ in the 2010s, meaning attacks were far rarer than expected in the 2000s and far more common in the 2010s. Severe weather runs the opposite direction ($+6.4$ in the 2000s, -4.7 in the 2010s) — proportionally displaced as attacks surged. Equipment failure also declined as a share ($+5.4$ in the 2000s, -4.0 in

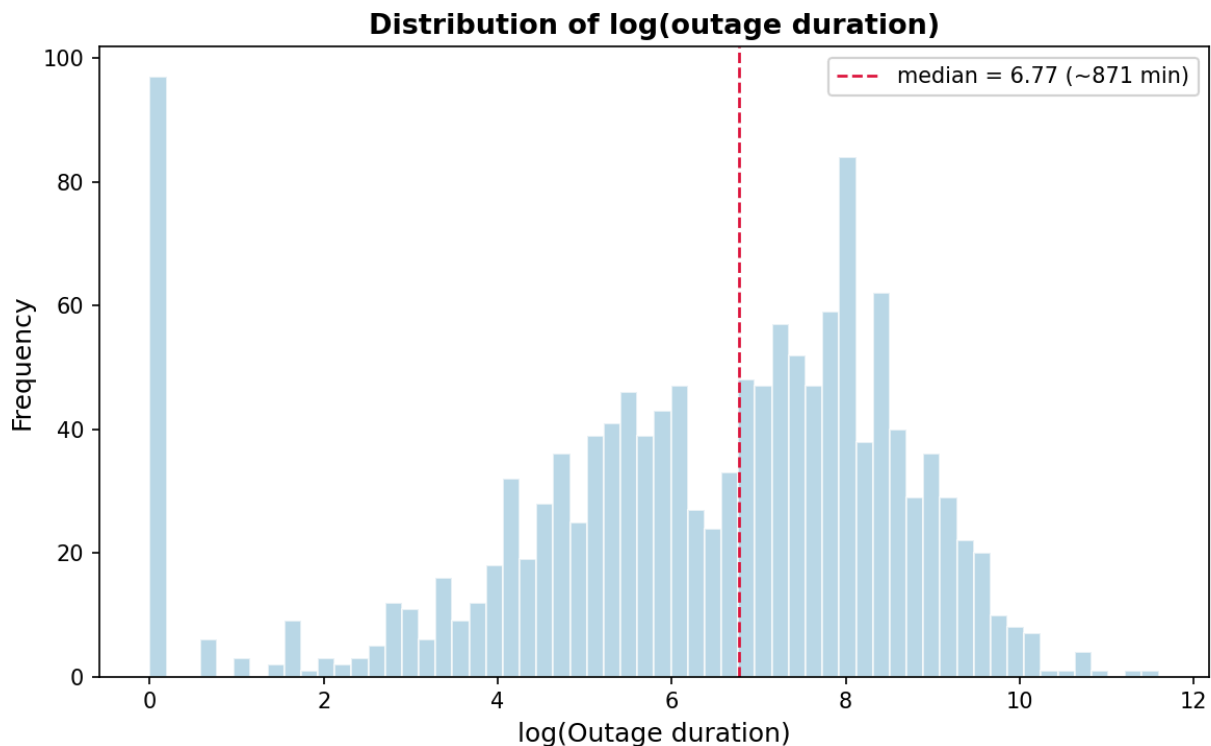
the 2010s). Fuel supply emergencies, islanding, and public appeal show residuals below $|2|$ in both decades, indicating their shares were stable across the period.

We see that the U.S. grid faced a fundamentally different threat profile in the 2010s than in the 2000s. Intentional attacks — near-absent before 2008 — became the second most common cause of major outages by the 2010s, reshaping the cause composition in a way that is statistically indistinguishable from zero probability under the null hypothesis of stability. This result directly connects to Test 1: if attack rates vary dramatically by decade and regions differ in their attack vulnerability, cause and region cannot be independent, which Test 1 confirmed.

5.3 MLE & LRT results

Summary of Results:

5.3.1) MLE distribution fitting summary (n = 1398 outages with positive duration):

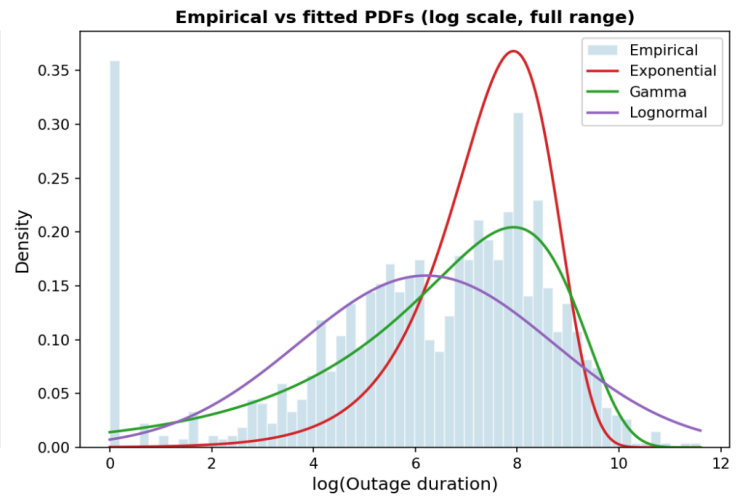
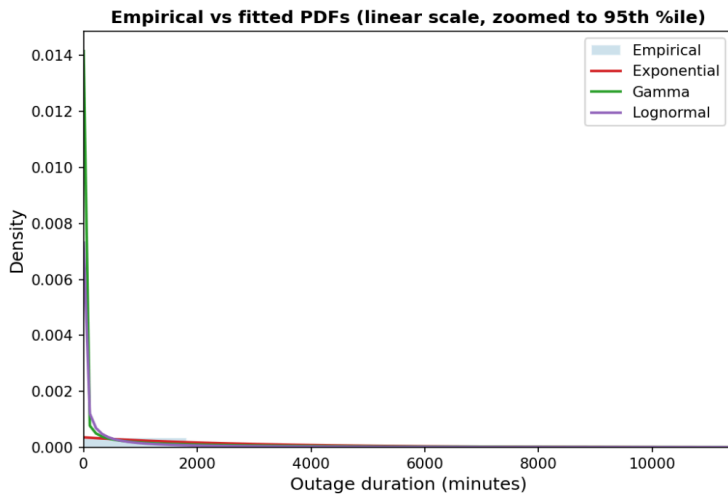


1. Summary statistics (minutes):

mean = 2771.9 median = 871.0 std = 6070.6 min = 1.0 max = 108653.0

2. MLE results (sorted by AIC):

Distribution	# of parameters	loglik	AIC	BIC	ΔAIC	ΔBIC
Gamma	2	-11826.04	23656.07	23666.56	0.00	0.00
Lognormal	2	-11945.96	23895.92	23906.40	239.84	239.84
Exponential	1	-12480.34	24962.68	24967.92	1306.60	1301.36



The Gamma distribution fits best both by AIC and BIC. The fitted parameters are as follows:

Exponential: scale = 2771.88; Gamma: shape = 0.3854, scale = 7192.61; Lognormal: sigma = 2.5008, mu = 6.2095.

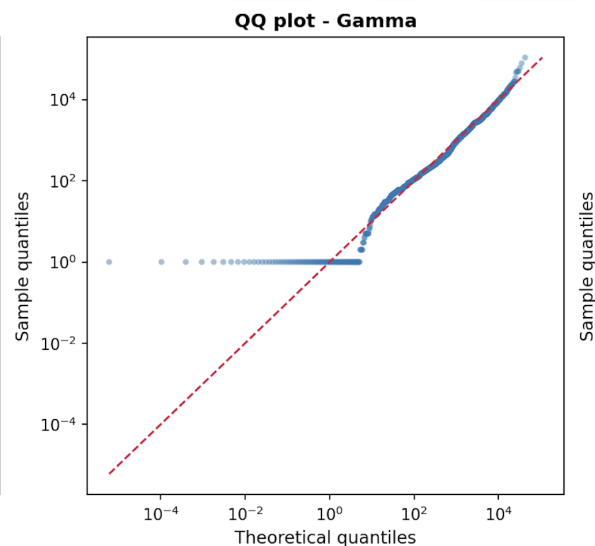
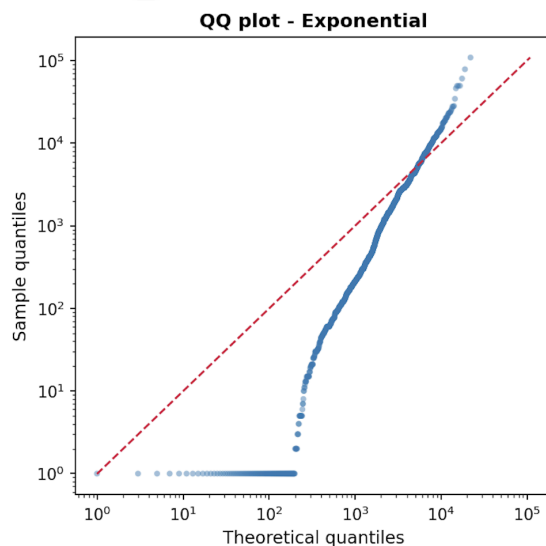
The Gamma distribution has a shape less than one, which signifies a decreasing hazard rate. Thus, this allows us to conclude that the longer an outage has lasted, the smaller the probability is of the outage ending at that moment. Most outages resolve within a short time period, but the remaining data points drag on for much longer; in turn, this produces a heavy tail.

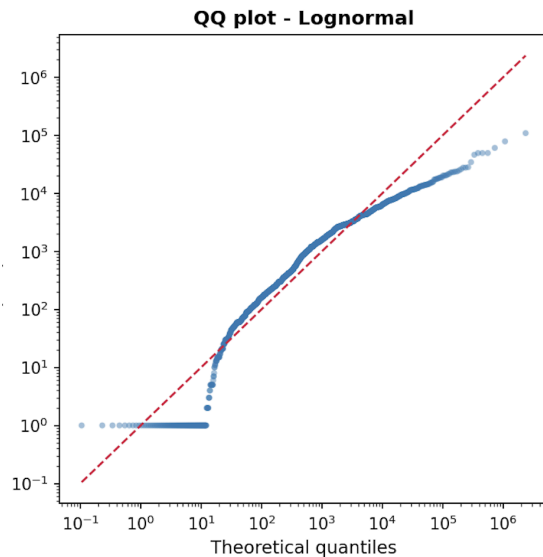
For the likelihood ratio test (Exponential vs. Gamma), our $X^2(1) = 1308.6028$, and our $p = 1.5265e-286$. Therefore, we reject the null hypothesis H_0 . The constant-rate Exponential distribution is inadequate at representing the data, thus confirming the need for a shape parameter and as a result the gamma distribution.

Finally, for our Kolmogorov-Smirnov goodness-of-fit tests:

Distribution	KS_statistic	p_vlaue
Exponential	0.2766	0.0000
Gamma	0.0462	0.0050
Lognormal	0.0926	0.0000

All three of the p-values fall under our significance level 0.05, so the K-S tests formally reject all distributions as an exact fit. Taking into account the sample size at $n = 1,398$, though, this makes sense as even tiny deviations are noticeable. Despite this, the Gamma distribution's KS-statistic is small enough to indicate a good and practical fit, although not exact.





5.3.2) Storm-severity Likelihood Ratio Test (n = 580)

1. Summary of results

The response variable was ``log(OUTAGE_DURATION)``.

mean = 7.433 sd = 1.528 Null model $\sigma = 1.528$ (logL -1,068.94)

full model $\sigma = 1.483$ (logL -1,051.65)

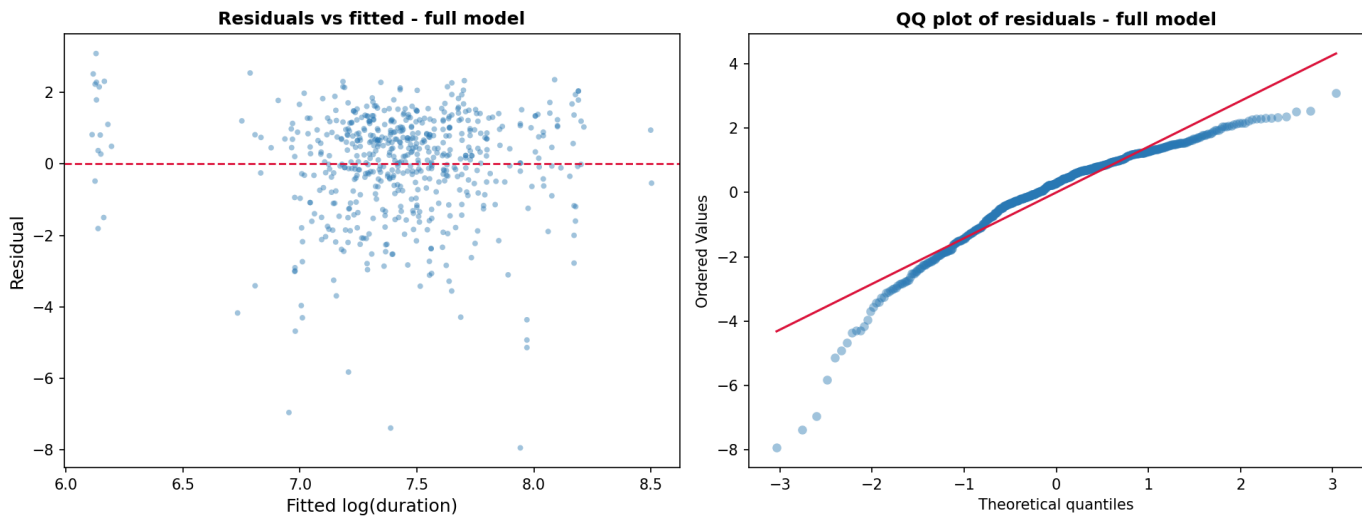
Full model coefficients:

Term	Estimate	Std. err	t	p-value
intercept	6.234	0.330	18.92	< 0.001
STORM_COUNT	0.00092	0.0014	0.68	0.499
MAX_MAGNITUDE	-0.00199	0.0042	-0.47	0.638
log(PROPERTY_DAMAGE)	0.0890	0.0176	5.06	<0.001
TOTAL_DEATHS	0.0025	0.0064	0.39	0.698

Out of the four recorded storm variables above, property damage is the only statistically significant effect. Due to the fact that it is on a log scale and that the coefficient's value is 0.0890, every ~2.7x increase in a storm's property damage is associated with 8.9% longer outage duration. The other three variables are not statistically significant enough to make any assumptions.

χ^2 When testing all four covariates together, we received the outputs (4) = 34.57, and our p-value = 5.7e-07. As we can see, the p-value is less than 0.05, thus we reject the null hypothesis and conclude that as a group, the storm covariates do in fact improve the model's fit, while property damage is the only sole significant variable.

The Pseudo- $R^2 = 0.058$, so the four storm covariates cover only roughly 6% of the variation in log-duration. Overall, we have concluded that storm severity is associated with longer outages, but much is still left to uncover since a majority of the variation, over 90%, is due to outside factors that were not in this test.



5.4 Nonparametric results

Test 1. KS, duration pre/post 2015. $D = 0.2314$, $p = 1.2e-06$. Pre-2015 median 840 min ($n = 1,323$); post-2015 median 220 min ($n = 153$).

We reject the null ($p = 1.2e-06$). The duration distributions differ between the two periods, and post-2015 outages are shorter. The post-2015 median of 220 minutes is about a quarter of the pre-2015 median of 840 minutes. This matches the regression in 5.1, where the YEAR coefficient was negative. Both point to durations falling over the study period.

Test 2. Wald-Wolfowitz runs test, monthly counts. Observed runs 54, expected runs 100.3, $Z = -6.59$, $p < 1e-6$.

We reject the null ($Z = -6.59$, $p < 1e-6$). There are far fewer runs than random ordering would give. Busy months and quiet months group into stretches instead of trading off month to month, which is what you would expect if weather seasons drive the counts.

Test 3. ACF, monthly counts. Lag 1 = 0.58, lag 6 = 0.34, lag 12 = 0.34, lag 24 = 0.28. The white-noise band is ± 0.139 , so lags 1, 6, and 12 are all significant and lag 24 is borderline.

Lag 1 is 0.58, so a busy month is usually followed by another busy month. Lag 12 sits at 0.34, above the band, which is the yearly cycle: counts repeat around the same months each year. The runs test and the ACF agree that outage frequency has time structure.

All three tests reject their null. Durations dropped after 2015, and monthly outage frequency is clustered and seasonal rather than random.

6. Discussion, Limitations & Conclusion

- Our three research questions can now be answered with the evidence assembled across the four analyses.
- First, have outage frequencies and durations increased over the study period? Frequency rose sharply: the population-normalized outage rate more than tripled between 2001 and 2011 (Fig 02), so the increase is real and not an artifact of population growth. Duration, by contrast, fell. The regression YEAR coefficient was negative (-0.049 , $p = 0.002$, about 4.8% shorter per year), and the Kolmogorov-Smirnov test independently confirmed the shift: the post-2015 median duration of 220 minutes is roughly a quarter of the pre-2015 median of 840 minutes ($D = 0.2314$, $p = 1.2e-06$). So outages became more frequent but shorter over the period.
- Second, which causes drive the trend? The chi-squared homogeneity test showed the cause composition changed strongly between the 2000s and 2010s ($\chi^2 = 343.3$, $p = 4.15e-71$, Cramér's $V = 0.47$), and the standardized residuals located the shift in intentional attacks, which were far rarer than expected in the 2000s and far more common in the 2010s. The regression explains why frequency and duration moved in opposite directions: attacks are the shortest-duration cause (92.5% shorter than severe weather), so as they came to dominate the event counts, they pulled the typical duration down even as the count of events climbed.
- Third, do severity distributions differ across regions? Yes. The chi-squared independence test rejected independence of sub-cause and NERC region ($\chi^2 = 524.5$, $p = 4.01e-66$), and the residuals showed regionally specialized threats: hurricanes in FRCC, wildfires and vandalism in WECC, thunderstorms in ECAR and RFC, winter weather in SERC. The regression added that two regions differ in restoration speed, with ECAR outages lasting about 110% longer than the RFC baseline and SERC about 48% shorter. The MLE work characterized the duration distribution itself, with the gamma fitting best (AIC 23,656 vs 23,896 for lognormal) and its shape parameter below one indicating a decreasing hazard rate, and the storm-severity LRT showed that storm covariates jointly add explanatory power ($\chi^2(4) = 34.57$, $p = 5.7e-07$), driven almost entirely by property damage.
- Several limitations bound these conclusions. The data ends in 2016, so we cannot speak to the last decade, and the apparent post-2014 decline in attacks is dataset truncation rather than a real reversal. The storm-severity test runs on only 580 merged events, and its pseudo- R^2 of about 0.06 means storm covariates explain a small share of duration variance, so much of what drives outage length is outside the variables we have. The data is observational, so every result is an association rather than a causal effect, and the complete-case handling of missing CUSTOMERS_AFFECTED (29% missing) and the collapsing of rare regions and sub-causes could bias some estimates. The regression also failed the Breusch-Pagan test for constant variance, so its borderline p-values should be read with caution.
- With more time and data, the clearest next steps are to extend the record past 2016 to test whether the frequency rise and duration decline continued, to model outage frequency directly with a Poisson or negative-binomial count model rather than treating duration alone, and to refit the regression with robust standard errors to address the heteroscedasticity. Even bounded as it is, the project gives a consistent picture: the U.S. grid has been hit more often but recovers faster, the cause mix shifted toward intentional attacks, and the threat profile is regional rather than national, so grid-hardening priorities should be set region by region.

7. Relation to Project Proposal

The main structure and objectives of our project from our proposal were upheld. We kept the same core-question - what drives major outages across the U.S. used the same datasets we proposed, and ran the same statistical methods that we had mentioned. Specifically, we ensured to conduct exploratory data analysis and regional trend visualization, regression with diagnostics, chi-squared tests, MLE distribution fitting, likelihood ratio tests, and the nonparametric tests in order to deduce a conclusion.

One thing that altered was a scope change in our data. We initially projected that we would want to use a time period like 2000-2024, but we pivoted to a smaller time frame. After diving deeper into the project, we realized the dataset had clean, working data only up until 2016. We decided to choose a decently well organized data set than a raw, messy recent dataset. Another thing that altered from the original plan was that sections like the storm-severity LRT used subsets of the dataset, like the weather-matched NOAA merge for necessary conclusions.

In terms of the feedback from the proposal, we have ensured to keep our conclusions and assumptions statistical, which we have attempted to do thoroughly in the Results & Analysis section. Although there was a comment about suggesting using Poisson or negative-binomial models, our project prioritized outage duration over frequency of events. Therefore we chose not to include these as other distributions aligned more.

8. References & Code

- Mukherjee, S., Nateghi, R., & Hastak, M. (2018). A multi-hazard approach to assess severe weather-induced major power outage risks in the U.S. *Reliability Engineering & System Safety*, 175, 283–305.
- NOAA Storm Events Database. National Centers for Environmental Information.
- Code: All analyses are reproducible from the submitted notebooks and script: 189FinalRegression.ipynb (regression and Model B LRT), chi_squared_v2.ipynb (independence and homogeneity tests), mle_analysis.ipynb and mle_lrt_storm.ipynb (distribution fitting and storm-severity LRT), nonparametric_tests.ipynb (KS, runs, and ACF), and eda_outages.py (exploratory data analysis and figures).